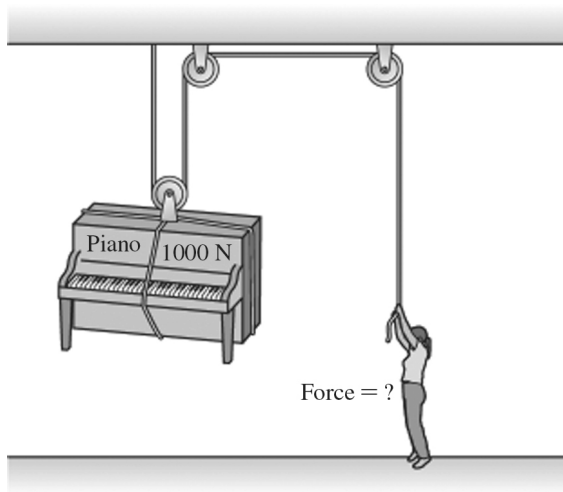


Name _____

1 minute = 60s. distance around a circle = $2\pi R$ weight = mass $\times$ g (on Earth $g = 9.8$) diameter = $2R$

- 1) A piano mover raises a 1000-N piano at a constant speed using a very light rope in a frictionless pulley system, as shown in the figure. With what force is the mover pulling down on the rope?

1) _____



- A) 250 N
 B) 500 N
 C) 1000 N
 D) 2000 N
 E) Depends on the speed of the piano.
- 2) A 10-kilogram block is pushed across a horizontal surface with a horizontal force of 20 N against a friction force of 10 N. The acceleration of the block is
- A) 2 m/s^2 .
 B) 5 m/s^2 .
 C) 10 m/s^2 .
 D) 1 m/s^2 .
 E) none of the above

2) _____

- 3) The components of vector $\vec{A}$ are given as follows:

3) _____

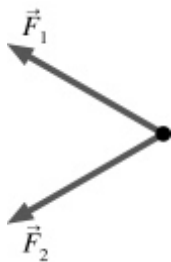
$$A_x = +6.1$$

$$A_y = -8.6$$

The angle measured counterclockwise from the x-axis to vector $\vec{A}$, in degrees, is closest to:

- A) 55 B) 305 C) 125 D) 145 E) 215

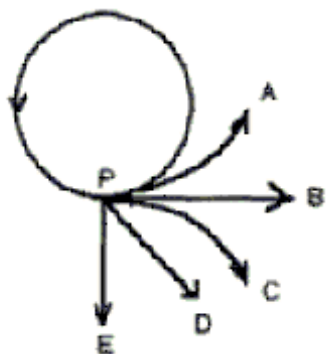
- 4) Calculate the impulse due to a force of 4.5 N that lasts for 1.4 s. 4) _____
 A) 5.0 kg·m/s B) 6.3 kg·m/s C) 5.7 kg·m/s D) 6.9 kg·m/s
- 5) A box with a weight of 100 N is at rest. It is then pulled by a 30 N horizontal force. Is this enough force to get it to move? ($\mu_k=0.30$, $\mu_s = 0.4$) 5) _____
 A) yes B) not enough information C) no
- 6) In which direction does the acceleration always face? 6) _____
 A) In the direction of the Net Force B) In the direction of the least applied force
 C) In the direction of the greatest applied force D) In the direction opposite of the Net Force
- 7) When a boxer is moving away from a punch, the force experienced is reduced because 7) _____
 A) all the above B) the time of contact is increased.
 C) momentum transfer is reduced. D) the force is less effective.
- 8) You swing a bat and hit a heavy box with a force of 1500 N. The force the box exerts on the bat is 8) _____
 A) greater than 1500 N if the bat bounces back.
 B) less than 1500 N if the box moves.
 C) greater than 1500 N if the box moves.
 D) exactly 1500 N whether or not the box moves.
 E) exactly 1500 N only if the box does not move.
- 9) A 10kg block is sliding along an inclined plane. the plane makes an angle of 20 degrees with the horizontal. The component of the weight along the plane (along the slope) is about 9) _____
 A) 100N B) 34N C) 0N D) 94N
- 10) A 0.2-kg rubber ball is dropped from the window of a building. It strikes the sidewalk below at 30m/s and rebounds up at 20m/s. The impulse on the ball during the collision is: 10) _____
 A) 2.0N · s downward E) 9.8N · s upward B) 2.0N · s upward
 C) 10N · s upward D) 10N · s downward
- 11) The figure shows two forces, each of magnitude 4.6 N, acting on an object. The angle between these forces is 40° , and they make equal angles above and below the horizontal. What third force will cause the object to be in equilibrium (acceleration equals zero)? 11) _____



- A) 7.0 N pointing to the right B) 4.3 N pointing to the right
 C) 3.5 N pointing to the right D) 8.6 N pointing to the right

- 12) A girl attaches a rock to a string, which she then swings counter-clockwise in a horizontal circle. The string breaks at point P in the figure, which shows a bird's-eye view (as seen from above). Which path (A-E) will the rock follow?

12) _____



A) path B E) path E
C) path D

B) path A
D) path C

- 13) If I weigh 741 N on Earth at a place where $g = 9.80 \text{ m/s}^2$ and 5320 N on the surface of another planet, what is the acceleration due to gravity on that planet?

13) _____

A) 61.2 m/s^2 B) 70.4 m/s^2 C) 51.4 m/s^2 D) 81.0 m/s^2

- 14) A stalled car is being pushed up a hill at constant velocity by three people. The net force on the car is

14) _____

A) down the hill and greater than the weight of the car.
B) up the hill and greater than the weight of the car.
C) down the hill and equal to the weight of the car.
D) zero.
E) up the hill and equal to the weight of the car.

- 15) The block is moving horizontally at a **constant velocity**. There are two applied forces on the object as shown in the image. In which direction is the friction force acting on the object?

15) _____

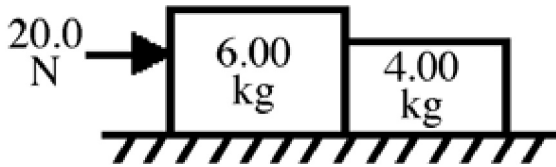


A) 7 N, rightward
C) 7 N, leftward E) no friction

B) 2 N, rightward
D) 2 N, leftward

- 16) A 6.00-kg block is in contact with a 4.00-kg block on a horizontal frictionless surface as shown in the figure. The 6.00-kg block is being pushed by a horizontal 20.0-N force as shown. What is the magnitude of the force that the 6.00-kg block exerts on the 4.00-kg block?

16) _____



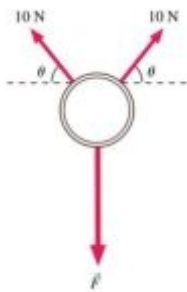
- A) 8.00 N B) 4.00 N C) 20.0 N D) 10.0 N E) 6.00 N
- 17) A 2.0-kg ball is moving with a constant speed of 5.0 m/s in a horizontal circle whose diameter is 1.0 m. What is the magnitude of the net force on the ball?
- A) 0 N B) 40 N C) 100 N D) 20 N E) 50 N
- 18) A bug lands on a wheel 0.2 m away from the center of rotation. If the wheel is turning at a rate of 120 rpm (revolution per minute), what is the velocity of the bug?
- A) 13.2m/s B) 50m/s
C) 5.36m/s D) 2.5 m/s E) 1.33m/s
- 19) An automobile and a golf cart traveling at the same speed collide head-on. The impact force is
- A) greater on the automobile.
B) greater on the golf cart.
C) the same for both.
- 20) This ring is in equilibrium and the angle theta is 40 degrees. What is the magnitude of F (round to nearest one)

17) _____

18) _____

19) _____

20) _____



- A) 13 N B) 8N E) 20N
C) 15N D) 6N

Answer Key

Testname: TEST2_FALL23

- 1) B
- 2) D
- 3) B
- 4) B
- 5) C
- 6) A
- 7) B
- 8) D
- 9) B
- 10) C
- 11) D
- 12) A
- 13) B
- 14) D
- 15) D
- 16) A
- 17) C
- 18) D
- 19) C
- 20) A